# *Supplement for* PrinGS: Privacy Preservation in Gaussian Splatting

This supplementary document provides additional implementation details in Section 1, our PrinGS-Humans benchmark details in Section 2, comparisons with trivial baselines in Section 4, and additional quantitative and qualitative results in Sections 5 and 7 that complement the experiments presented in the main paper. Finally, we address some limitations of our method in Section 8.

## 1 Implementation Details

All experiments are built on top of the official 3DGS implementation on a single NVIDIA RTX 3090 Ti GPU with 24 GB VRAM. Every eighth frame is reserved for evaluation while the remaining frames are used during optimization, following standard protocol.

### 1.1 Preprocessing

For obtaining camera poses and sparse point clouds, COLMAP is used across all datasets. Both the post-training and privacy-by-design frameworks require detecting sensitive regions in images. In both cases, SAM3 is leveraged alongside the text prompts specified in Table 1 for masking.

The resulting output is a binary mask $M_v$ where white pixels denote sensitive regions ($M_v(p) = 1$) and black pixels denote otherwise ($M_v(p) = 0$). For the Truck and M60 scenes, morphological erosion is additionally applied to prevent masks from erroneously covering background Gaussians that do not contribute to license plates or identification numbers. Experiments indicate that such morphological adjustments are not required for other datasets.

The final set of utilized datasets, including COLMAP outputs and binary masks, will be released upon publication.

### 1.2 3D Segmentation of Sensitive Gaussians

As described in Section 3.1 of the main paper, sensitive Gaussians are identified by tracking their contributions during alpha compositing. The transmittance threshold is set to $\tau_T = 0.05$, ensuring that accumulated contributions significantly affect color during the alpha blending stage. For comparison, the standard 3D Gaussian Splatting rasterizer accumulates color up to a transmittance of 0.01, making this threshold slightly stricter to avoid noisy allocations. The transmittance evaluation ($T_i$) follows the formulation from the original 3D Gaussian Splatting rasterizer.

To eliminate spurious background floaters that coincidentally project into the two-dimensional masks, the flagged Gaussians are filtered using an Isolation Forest. Specifically, if the count of contributing indices is small (fewer than ten), they are retained directly without filtering. Otherwise, an Isolation Forest with a contamination parameter of 0.1 is fitted on the spatial means $\mu_i$ of the sensitive Gaussians. For multi-view aggregation across views, the global visibility threshold is set to $\tau_v = 1$, requiring a Gaussian to be flagged as sensitive in at least one view to be considered for targeted anonymization.

**Table 1: Text prompts used with SAM3 across different evaluation datasets to generate binary privacy masks.**

| Dataset | SAM3 Text Prompts |
|---|---|
| House | "human face", "eyebrows", "lips", "upper lips", "lower lips", "beard" |
| Hospital | "human face", "eyebrows", "lips", "upper lips", "lower lips", "beard" |
| Truck | "vehicle license plate" |
| Barn | "house number 2150 on wooden beam" |
| M60 | "rectangular bumper number", "identification marking on tank" |
| Piano | "brand name on piano" |
| Car | "vehicle license plate" |

### 1.3 Gaussian Anonymization Formulation

As detailed in Section 3.2 of the main paper, once the sensitive subset $\mathcal{S} \subset \mathcal{G}$ is isolated, localized volumetric blurring is applied via $k$-nearest neighbor neighborhoods. For facial regions, the weight function $f(s_j)$ incorporates scale-dependent weighting with a weight power of 2. Specifically, to reduce the three-dimensional anisotropic scale vector of each neighboring Gaussian (represented along its $x$, $y$, and $z$ axes) into a single scalar number, the mean across the spatial scale dimensions is computed to obtain a representative average scale value. This isotropic scale value is then raised to the specified weight power. This ensures that larger background structures appropriately guide the color smoothing of facial features while preserving the underlying head geometry.

### 1.4 Post-Training Privacy Preservation

Unlike standard training pipelines, the post-training framework operates on an already optimized 3D Gaussian Splat. A key prerequisite is utilizing the SIBR viewer to store camera frames in a keypoint format that is subsequently consumed by the pipeline.

### 1.5 Privacy-by-Design

In contrast to the post-training setting, the privacy-by-design framework integrates directly into the training loop. For facial regions, for saving at an intermediate iteration, such as 7,000, a copy of the current splat optimized via standard 3D Gaussian Splatting is instantiated, and single-view segmentation followed by anonymization operators is applied. Crucially, any ongoing optimization beyond that iteration proceeds using the standard un-anonymized 3D Gaussian Splatting optimization stream rather than the branched anonymized copy. Conversely, for text regions, at designated milestone iterations, such as iterations 7,000, 14,000, 21,000, and 30,000



**Figure 1: Overview of the PrinGS-Humans synthetic benchmark scenes. Left: Sample rendered views from the House (top) and Hospital (bottom) scenes. Right: Corresponding camera trajectories that spans the scene.**

in the experiments, the active optimization state of the splat is directly edited to apply multi-view segmentation and anonymization inline, without maintaining a separate static copy.

### 1.6 Evaluation Protocol

To quantitatively assess privacy protection and visual utility, we employ recall-based metrics for both facial similarity and text recognition. Specifically, facial similarity is evaluated using the Face Similarity Measure (FSM) via the InsightFace framework (specifically the *buffalo_l* model pack), while text recognition is evaluated using the Character Error Rate (CER) via Google OCR from the Cloud Vision API. Inspired by STRinGS, both metrics focus on recall-based evaluation to rigorously measure residual sensitive information.

Because the two-dimensional face detection process can introduce localized spatial shifts, we perform Intersection-over-Union (IoU) based overlap to match face detections between the rendered and ground-truth images, followed by computing the cosine similarity between their embeddings. In cases where a detection in the ground-truth image does not have a corresponding detection in the rendered image, the similarity is treated as 0.

This convention makes FSM a recall-based privacy metric, but it also means that FSM in isolation rewards a method that destroys the face entirely rather than anonymizing it: with no detection in the render, the similarity is 0 by construction. We therefore pair FSM with the Face Detection Rate (FDR), defined as the fraction of ground-truth face instances across the held-out evaluation views for which the SCRFD-10GF detector returns a spatially matching detection. FDR is reported as a utility metric. Because the InsightFace detector reasons jointly over facial appearance and facial geometry, FDR remains high when the head structure and silhouette survive anonymization and collapses towards zero when they do not. A retained FDR in the 36–43% range, as achieved by PrinGS, therefore reflects preserved head geometry with suppressed fine-grained facial detail, which is exactly the intended behaviour of representation-level anonymization.

For text evaluation, OCR detection frequently suffers from text fragmentation, making strict one-to-one string comparisons insufficient. Following STRinGS, we construct a bipartite graph where nodes represent OCR-detected text regions and edges connect regions with an IoU exceeding 0.1. Connected components yield robust many-to-many matches that naturally handle fragmentation (such as 1-to-$N$ or $N$-to-1 splits), avoiding the rigid constraints of traditional Hungarian matching. Any ground-truth text entirely undetected in the render is treated as a complete loss of characters.

## 2 PrinGS-Humans Benchmark

Due to inherent privacy constraints, public benchmarks featuring static indoor scenes with human subjects are practically nonexistent. To enable a rigorous and controlled evaluation of facial privacy preservation, we introduce a custom synthetic benchmark constructed entirely within Blender.

Camera trajectories are meticulously planned to provide dense coverage across the entire scene, ensuring that every subject is observed from multiple distinct viewpoints. Representative visual examples of the synthetic benchmark are illustrated in Figure 1.

Figure 2 presents representative views of all nine subjects in our benchmark, comprising five individuals in House and four in Hospital. The subjects exhibit diverse appearances and facial characteristics, enabling evaluation of privacy preservation across a heterogeneous set of identities. Each subject is observed from multiple viewpoints to assess the consistency of anonymization across rendered views.

## 3 Per-Subject Face Privacy Results

Table 2 reports FSM for every subject in the benchmark. Because subjects appear across a varying number of evaluation frames,




**Figure 2: Human subjects in the PrinGS-Humans benchmark. Representative views of all nine subjects across the House (P1–P5) and Hospital (P1–P4) scenes, illustrating diversity in facial appearance, skin tone, gender, clothing, and pose.**

**Table 2: Per-subject Face Similarity Measure (FSM) on our PrinGS-Humans benchmark. Lower values indicate stronger anonymization. Weighted averages over all occurrences (#Occ) yield the scene-level FSM reported in Table 1 of the main paper.**

| FSM ↓ | Person | #Occ | 3DGS | **Ours** |
|---|---|---|---|---|
| House | P1 | 7 | 86.71 | 10.12 |
| | P2 | 8 | 85.62 | 0.00 |
| | P3 | 7 | 79.86 | 8.86 |
| | P4 | 5 | 52.60 | 10.20 |
| | P5 | 9 | 79.89 | 5.78 |
| | Weighted Average | | 78.68 | 6.55 |
| Hospital | P1 | 10 | 89.60 | 10.90 |
| | P2 | 10 | 90.30 | 19.90 |
| | P3 | 9 | 91.11 | 19.89 |
| | P4 | 11 | 83.91 | 1.91 |
| | Weighted Average | | 88.55 | 12.70 |

per-subject values are aggregated via weighted averages over all occurrences to obtain the scene-level FSM reported in the main paper. Standard 3DGS yields high similarity for every subject, with a minimum of 52.60% on House and 83.91% on Hospital, whereas our method reduces the worst case to 10.20% on House and 19.90% on Hospital, with a best case of 0.00%. A residual FSM of roughly 10–20% for an individual does not reflect a failure to anonymize, but rather captures the embedding-space similarity that remains as a consequence of preserving natural facial structural coherence, which is the same property that keeps FDR non-zero.

## 4 Trivial Baselines

No direct prior baselines exist for our post-training scenario, which is uniquely application-focused and yields anonymized splats within one second. For the privacy-by-design setting, however, three trivial baselines can be constructed. We evaluate all of them against standard 3DGS and PrinGS, reporting the privacy–utility trade-off on the face scenes in Table 3, OCR-CER on the text scenes in Table 4, and rendering fidelity in Table 5.

### 4.1 B1: Input-Level Anonymization Before Preprocessing

Sensitive regions are blurred directly in the input images before any preprocessing, after which the full pipeline, Structure-from-Motion followed by 3DGS optimization, is run on the anonymized images. Where sensitive regions occupy a large part of the scene, this introduces a severe loss of the discriminative local image patterns required by feature-matching algorithms such as SIFT, substantially reducing the robustness of COLMAP and resulting in incomplete camera registration. On both PrinGS-Humans scenes, where faces are prominent, COLMAP fails to register the scene entirely and no reconstruction is obtained, which we report as – in all tables. Even on the text scenes, where the sensitive region covers only a small portion of each image and registration succeeds, B1 is the weakest method in fidelity, losing 2.9 dB PSNR on Truck and 8.3 dB on Piano. Its privacy guarantee is also the weakest: because a static blur kernel is applied independently in each view, the effective blur strength varies with viewing distance and the text remains partially legible from near views, yielding an OCR-CER of only 0.901–0.917 on Barn, M60, and Piano against the 1.00 achieved by PrinGS.

### 4.2 B2: COLMAP on Original Images with Blurry Optimization

COLMAP is run on the original, uncompromised images to obtain camera poses, but the subsequent optimization is supervised using blurred images. This sidesteps the registration failure of B1 and preserves rendering fidelity, but suffers from two primary issues:

- *View-dependent inconsistencies:* The apparent size and orientation of sensitive regions vary significantly across viewpoints. Consequently, applying a static blur kernel leads to inconsistent observations across frames, introducing noticeable artifacts into the reconstructed scene.
- *Geometry degradation:* Eliminating image details prior to reconstruction removes critical geometric cues around sensitive regions (such as facial contours and silhouettes), leading to degraded geometry and compromised scene fidelity.

The consequence is visible in Table 3: B2 attains an FSM of 0.00 on both scenes, but only because the detector no longer finds a face anywhere in the render, giving an FDR of 0.00. This is destruction rather than anonymization, and is precisely the failure mode that FDR is designed to expose.

### 4.3 B3: Pruning the Sensitive Gaussians

The sensitive Gaussians are localized exactly as in PrinGS, but instead of being anonymized they are removed from the representation, equivalently by zeroing their opacity. Deleting the subset

**Table 3: Privacy–utility trade-off against the trivial baselines on the two PrinGS-Humans scenes. FDR (utility) measures the fraction of ground-truth faces that remain detectable in the anonymized render, where higher is better, while FSM (privacy) measures residual identity leakage, where lower is better. B1 blurs the input images before preprocessing, B2 uses clean COLMAP poses with blurred optimization, and B3 prunes the sensitive Gaussians. B1 is reported as – because COLMAP fails to register these scenes. B2 and B3 reach a low FSM only by destroying the face, driving FDR towards 0, whereas PrinGS is the only method that suppresses identity while keeping the face detectable.**

| | House | | Hospital | |
|---|---|---|---|---|
| Method | FDR ↑ | FSM ↓ | FDR ↑ | FSM ↓ |
| 3DGS | 91.67 | 78.68 | 97.50 | 88.55 |
| B1 | – | – | – | – |
| B2 | 0.00 | 0.00 | 0.00 | 0.00 |
| B3 | 11.11 | 0.20 | 15.00 | 1.22 |
| **Ours** | **36.11** | 6.55 | **42.50** | 12.70 |

**Table 4: OCR Character Error Rate (OCR-CER) against the trivial baselines on the text scenes. Higher values indicate stronger anonymization, with 1.00 corresponding to complete failure of text recognition. B1 leaks text from near viewpoints because the static per-view blur kernel is applied inconsistently across viewing distances, and B3 leaves the Piano brand name partially legible. PrinGS is the only method that attains the maximum OCR-CER on every scene.**

| OCR-CER ↑ | Truck | Barn | M60 | Piano | Car |
|---|---|---|---|---|---|
| 3DGS | 0.500 | 0.028 | 0.207 | 0.149 | 0.158 |
| B1 | 1.000 | 0.917 | 0.901 | 0.914 | 1.000 |
| B2 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| B3 | 1.000 | 1.000 | 1.000 | 0.954 | 1.000 |
| **Ours** | **1.000** | **1.000** | **1.000** | **1.000** | **1.000** |

leaves a hole that the surrounding Gaussians must fill, costing 3.2 dB PSNR on House and 4.6 dB on Hospital, the largest fidelity drop of any method on the face scenes, and producing visible artifacts around the excised region. Its residual FDR of 11.11–15.00% at an FSM of 0.20–1.22% again reflects a largely destroyed face rather than a preserved but anonymous one.

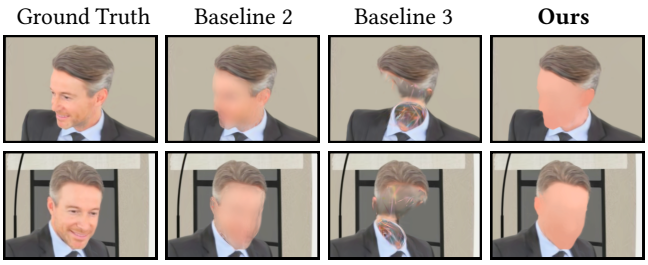



**Figure 3: Qualitative comparison of privacy-by-design facial anonymization against trivial baselines on the PrinGS-Humans dataset. From left to right: ground truth, B2 (optimization using blurred images with clean COLMAP poses), B3 (pruning sensitive Gaussians), and PrinGS (ours). B2 suppresses facial features at the cost of degraded facial geometry, while B3 introduces severe structural artifacts by removing sensitive Gaussians. In contrast, PrinGS effectively obscures facial identity while preserving coherent head geometry and scene appearance across viewpoints.**

Representative qualitative comparisons against the baselines are presented in Figure 3. Overall, PrinGS is the only method that drives FSM down while keeping a substantial fraction of the faces detectable, and it does so at a fidelity cost of under 1.2 dB PSNR on every scene except Car, where all methods including the baselines drop by a comparable 2.1–2.5 dB. This follows from modifying only the appearance parameters of the sensitive Gaussians while leaving their positions, scales, and opacities untouched, which decouples identity from physical structure.

## 5 Rendering Quality Metrics

The primary objective of our method is privacy preservation rather than maximizing rendering fidelity. Nevertheless, it remains important to ensure that anonymization minimally affects the visual appearance of the remaining background scene. Consequently, conventional rendering quality metrics, including PSNR, SSIM, and LPIPS, are additionally reported for PrinGS, standard 3DGS, and the three trivial baselines of Section 4.

In privacy-preserving scenarios, because anonymized renders intentionally obscure fine-grained facial and textual details present in the original ground-truth images, standard error-based metrics naturally yield lower quantitative scores (such as reduced PSNR and SSIM, and increased LPIPS) compared to standard 3DGS.

Table 5 compares our approach against standard 3D Gaussian Splatting and the three baselines. The results demonstrate that while our method incurs the expected degradation in PSNR and SSIM alongside an increase in LPIPS due to intentional anonymization, the overall visual quality and fidelity of the background scene are well-preserved without catastrophic degradation.

## 6 Post-Training Quantitative Evaluation

The post-training setting is the headline application of PrinGS: it assumes that only a trained splat is available and that the original calibrated training images are inaccessible. Under that assumption no ground truth exists against which renders could be scored, which is why the main paper presents this setting qualitatively.

To nevertheless quantify it, we construct a controlled protocol in which the training images are assumed available *purely for evaluation purposes* and are never used to retrain the splat. Specifically, a vanilla 3DGS splat is first optimized as usual. Our post-training anonymization is then applied to this trained splat using target frames containing the sensitive regions as the user-provided reference views, exactly as a user would do interactively in the SIBR viewer. Finally, the anonymized splat is rendered on the held-out evaluation views, that is every eighth frame, and privacy leakage is measured using the same protocol as in the privacy-by-design setting of the main paper.

Table 6 reports the resulting metrics on the face scenes. The post-training mode attains an FSM of 7.00 on House and 14.20 on Hospital, closely matching the 6.55 and 12.70 obtained by the privacy-by-design variant, at rendering quality that is likewise comparable to both the privacy-by-design variant and standard 3DGS. On all five text scenes, the post-training mode attains an OCR-CER of 1.000, identical to the privacy-by-design setting. These results confirm that operating directly on an already-trained representation sacrifices neither privacy nor fidelity relative to anonymizing during optimization, while requiring under one second and no access to the original training data.

## 7 Additional Qualitative Results

### 7.1 Privacy-by-Design

Figure 4 presents additional qualitative comparisons between standard 3D Gaussian Splatting, and our method. Across both facial and textual privacy scenarios, our approach consistently suppresses

**Table 5: Comparison of rendering quality metrics across standard 3DGS, PrinGS, and the three trivial baselines (B1: input-level blur before preprocessing, B2: clean COLMAP poses with blurred optimization, B3: pruning the sensitive Gaussians). Higher PSNR and SSIM indicate better rendering fidelity, while lower LPIPS indicates improved perceptual similarity. It is but natural that due to anonymization, these metrics which measure scene fidelity will be degraded for every anonymizing method. For PrinGS the difference with respect to 3DGS is only marginal, staying within 1.2 dB PSNR on all scenes except Car, where every method drops comparably, indicating the scene quality is not compromised during optimization. Entries marked – correspond to B1 on the House and Hospital scenes, where blurring the input images causes COLMAP to fail to register the scene entirely.**

| | PSNR ↑ | | | | | SSIM ↑ | | | | | LPIPS ↓ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Scene | 3DGS | **Ours** | B1 | B2 | B3 | 3DGS | **Ours** | B1 | B2 | B3 | 3DGS | **Ours** | B1 | B2 | B3 |
| House | 37.94 | 37.05 | – | 37.47 | 34.73 | 0.9616 | 0.9599 | – | 0.9603 | 0.9577 | 0.1066 | 0.1095 | – | 0.1101 | 0.1117 |
| Hospital | 39.91 | 38.80 | – | 39.76 | 35.27 | 0.9749 | 0.9735 | – | 0.9742 | 0.9709 | 0.1256 | 0.1289 | – | 0.1278 | 0.1326 |
| Truck | 25.49 | 25.07 | 22.62 | 25.48 | 24.39 | 0.8850 | 0.8790 | 0.7680 | 0.8844 | 0.8716 | 0.1420 | 0.1484 | 0.2080 | 0.1431 | 0.1554 |
| Barn | 28.43 | 28.38 | 28.39 | 28.43 | 28.48 | 0.8598 | 0.8606 | 0.8597 | 0.8597 | 0.8601 | 0.2161 | 0.2154 | 0.2164 | 0.2159 | 0.2160 |
| M60 | 28.00 | 28.02 | 28.12 | 28.11 | 26.92 | 0.9033 | 0.9028 | 0.9037 | 0.9036 | 0.8962 | 0.1739 | 0.1755 | 0.1731 | 0.1743 | 0.1802 |
| Piano | 30.33 | 30.34 | 22.05 | 30.41 | 30.34 | 0.9231 | 0.9223 | 0.7883 | 0.9228 | 0.9226 | 0.2424 | 0.2437 | 0.3408 | 0.2432 | 0.2446 |
| Car | 36.68 | 34.56 | 34.65 | 34.45 | 34.21 | 0.9618 | 0.9619 | 0.9618 | 0.9616 | 0.9614 | 0.0723 | 0.0718 | 0.0721 | 0.0718 | 0.0727 |

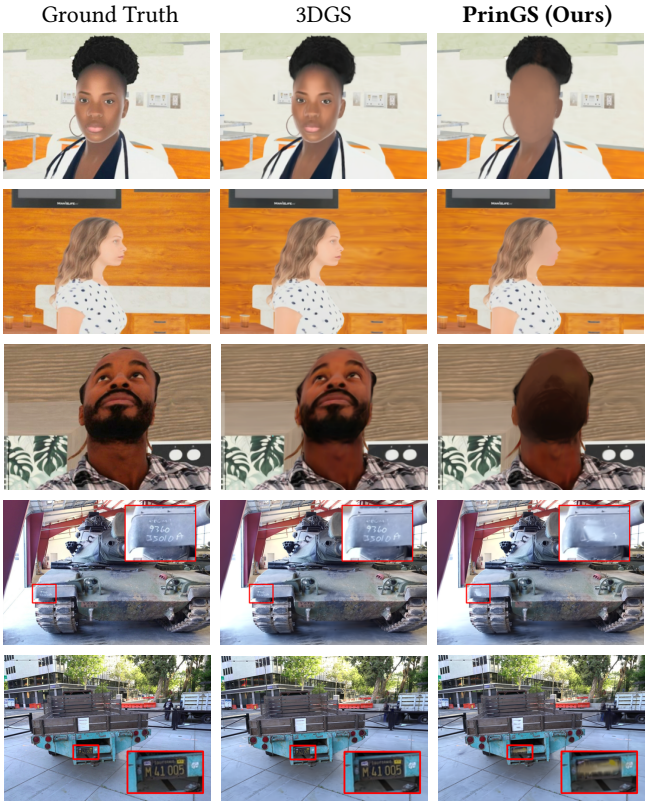



**Figure 4: Qualitative comparison of privacy-by-design training on scenes from our PrinGS-Humans dataset (rows 1-3) and Tanks and Temples (rows 4,5). From left to right: ground truth image, novel rendered images from standard 3D Gaussian Splatting (3DGS), and our method PrinGS. While standard 3DGS reconstructs sensitive faces and text, our method prevents them from being encoded during training, preserving the appearance and geometry of the remaining scene.**

**Table 6: Quantitative evaluation of post-training anonymization on held-out evaluation views. The anonymization is applied directly to an already-trained vanilla 3DGS splat, with no retraining and no access to the training images during anonymization. The resulting scores closely match those reported for the privacy-by-design setting in Table 1 of the main paper, confirming that the headline post-training mode achieves comparable privacy at comparable fidelity.**

| Scene | PSNR ↑ | SSIM ↑ | LPIPS ↓ | FSM ↓ |
|---|---|---|---|---|
| House | 37.34 | 0.9602 | 0.1094 | 7.00 |
| Hospital | 39.48 | 0.9742 | 0.1276 | 14.20 |

sensitive information while preserving the appearance of the remaining scene.

## 7.2 Post-training Anonymization

Figure 5 presents additional examples of our post-training anonymization framework. Given a single user-selected reference view, the learned Gaussian representation is edited directly, resulting in viewpoint-consistent anonymization across all rendered novel views.

## 8 Limitations

Our current framework assumes that sensitive regions are specified either through user interaction on a reference view or using automated segmentation models like SAM3 combined with preprocessing techniques such as erosion. Consequently, our three-dimensional sensitive region detections are inherently constrained by the quality of the binary masks produced by these models. Furthermore, while this work primarily focuses on facial and textual privacy, the proposed framework is general and can potentially be extended to other forms of sensitive information such as computer screens, personal belongings, and biometric identifiers, by adapting and experimenting with our anonymization operators.




**Figure 5: Qualitative results for post-training anonymization on scenes from our PrinGS-Humans dataset (rows 1-3) and Tanks and Temples (row 4). Starting from a single user-selected *reference view* (left), our method directly anonymizes the underlying Gaussian representation without requiring access to the original training images. The remaining columns show novel views rendered after representation-level anonymization. Sensitive facial and textual information is consistently removed across arbitrary viewpoints, demonstrating that privacy preservation naturally propagates through the edited 3D representation rather than being limited to the selected view.**